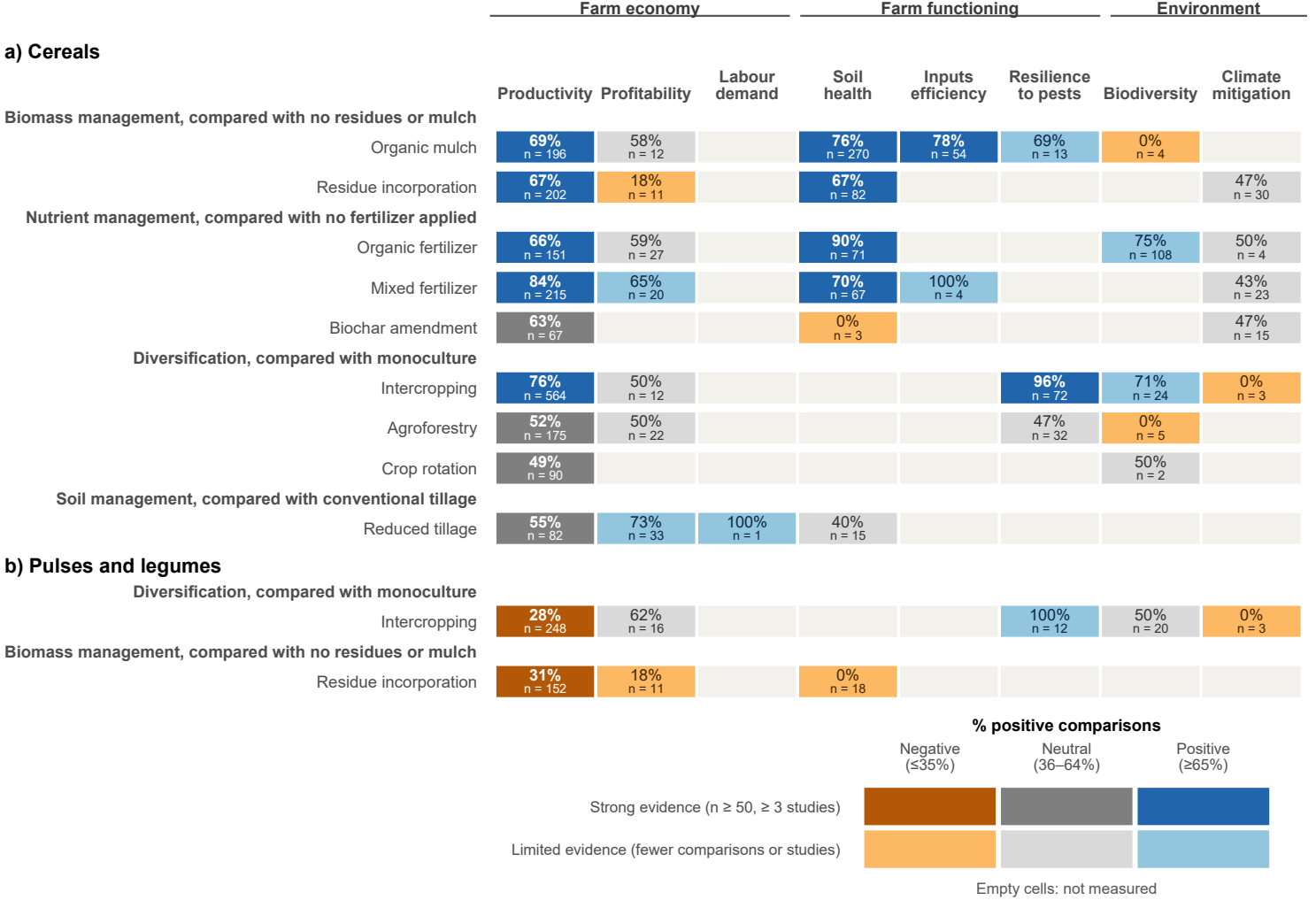
Farm economy
Farm functioning
Environment
a) Cereals
Productivity
Profitability
Labour demand
Soil health
Inputs efficiency
Resilience to pests
Biodiversity
Climate mitigation
Biomass management, compared with no residues or mulch
Organic mulch
69% n = 196
58% n = 12
76% n = 270
78% n = 54
69% n = 13
0% n = 4
Residue incorporation
67% n = 202
18% n = 11
67% n = 82
47% n = 30
Nutrient management, compared with no fertilizer applied
Organic fertilizer
66% n = 151
59% n = 27
90% n = 71
75% n = 108
50% n = 4
Mixed fertilizer
84% n = 215
65% n = 20
70% n = 67
100% n = 4
43% n = 23
Biochar amendment
63% n = 67
0% n = 3
47% n = 15
Diversification, compared with monoculture
Intercropping
76% n = 564
50% n = 12
96% n = 72
71% n = 24
0% n = 3
Agroforestry
52% n = 175
50% n = 22
47% n = 32
0% n = 5
Crop rotation
49% n = 90
50% n = 2
Soil management, compared with conventional tillage
Reduced tillage
55% n = 82
73% n = 33
100% n = 1
40% n = 15
b) Pulses and legumes
Diversification, compared with monoculture
Intercropping
28% n = 248
62% n = 16
100% n = 12
50% n = 20
0% n = 3
Biomass management, compared with no residues or mulch
Residue incorporation
31% n = 152
18% n = 11
0% n = 18
% positive comparisons
Negative (≤35%)
Neutral (36–64%)
Positive (≥65%)
Strong evidence (n ≥ 50, ≥ 3 studies)
Limited evidence (fewer comparisons or studies)
Empty cells: not measured